3.2.1. Numpy: Matrix Operations

05:51

The given code takes two matrices, , and , as input from the user and converts them into NumPy arrays.

Task:
You are required to compute and display the results of the following matrix operations:

1. **Addition** ()
2. **Subtraction** ()
3. **Element-wise Multiplication** ()
4. **Matrix Multiplication** ()
5. **Transpose of Matrix A**

Input Format:

- The user will input 3 rows for , each containing 3 integers separated by spaces.
- Similarly, the user will input 3 rows for , each containing 3 integers separated by spaces.

Output Format:
The program should display the results of the operations in the following order:

1. The result of Addition.
2. The result of Subtraction.
3. The result of Element-wise Multiplication.
4. The result of Matrix Multiplication.
5. The Transpose of Matrix A.

Code:

```
import numpy as np



# Input matrices

print("Enter Matrix A:")

matrix_a = np.array([list(map(int, input().split())) for i in range(3)])



print("Enter Matrix B:")
```

```
matrix_b = np.array([list(map(int, input().split())) for i in range(3)])



# Addition
maddition=matrix_a + matrix_b
print("Addition (A + B):")
print(maddition)
# Subtraction
msubtraction=matrix_a - matrix_b
print("Subtraction (A - B):")
print(msubtraction)
# Multiplication (element-wise)
mmultiplication=matrix_a * matrix_b
print("Element-wise Multiplication (A * B):")
print(mmultiplication)
# Matrix multiplication (dot product)
mdp=np.dot(matrix_a,matrix_b)
print("A dot B:")
print(mdp)
# Transpose
mtrsp=matrix_a.T
print("Transpose of A:")
print(mtrsp)
```

Explorer

Average time	Maximum time
0.078 s	0.093 s
77.75 ms	93.00 ms

2 out of 2 shown test case(s) passed

2 out of 2 hidden test case(s) passed

Test case 1 93 ms Debug

Expected output	Actual output
Enter Matrix A:	Enter Matrix A:
1 2 3	1 2 3
4 5 6	4 5 6
7 8 9	7 8 9
Enter Matrix B:	Enter Matrix B:
1 1 1	1 1 1
1 1 1	1 1 1
1 1 1	1 1 1
Addition (A + B):	Addition (A + B):
[[2 3 4]	[[2 3 4]
[5 6 7]	[5 6 7]
[8 9 10]]	[8 9 10]]
Subtraction (A - B):	Subtraction (A - B):
[[0 1 2]	[[0 1 2]
[3 4 5]	[3 4 5]
[6 7 8]]	[6 7 8]]
Element-wise Multiplication (A * B):	Element-wise Multiplication (A * B):
[[1 2 3]	[[1 2 3]
[4 5 6]	[4 5 6]
[7 8 9]]	[7 8 9]]
A dot B:	A dot B:
[[6 6 6]	[[6 6 6]
[15 15 15]	[15 15 15]
[24 24 24]]	[24 24 24]]
Transpose of A:	Transpose of A:
[[1 4 7]	[[1 4 7]

Terminal Test cases

Average time	Maximum time
0.078 s	**0.093 s**
77.75 ms	93.00 ms

2 out of 2 shown test case(s) passed

2 out of 2 hidden test case(s) passed

Test case 2 73 ms Debug

Expected output	Actual output
Enter Matrix A:	Enter Matrix A:
2 4 8	2 4 8
9 6 5	9 6 5
7 8 9	7 8 9
Enter Matrix B:	Enter Matrix B:
10 12 13	10 12 13
14 15 16	14 15 16
17 18 19	17 18 19
Addition (A + B):	Addition (A + B):
[[12 16 21]	[[12 16 21]
[23 21 21]	[23 21 21]
[24 26 28]]	[24 26 28]]
Subtraction (A - B):	Subtraction (A - B):
[[-8 -8 -5]	[[-8 -8 -5]
[-5 -9 -11]	[-5 -9 -11]
[-10 -10 -10]]	[-10 -10 -10]]
Element-wise Multiplication (A * B):	Element-wise Multiplication (A * B):
[[20 48 104]	[[20 48 104]
[126 90 80]	[126 90 80]
[119 144 171]]	[119 144 171]]
A dot B:	A dot B:
[[212 228 242]	[[212 228 242]
[259 288 308]	[259 288 308]
[335 366 390]]	[335 366 390]]
Transpose of A:	Transpose of A:
[[2 9 7]	[[2 9 7]
[4 6 8]	[4 6 8]